

Benefit analysis of shared depot resources for multi-depot vehicle routing problem with fuel consumption

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ABSTRACT

Shared depot resources, which allow the route of a vehicle to start from a depot and end at any one of the depot sets, can potentially reduce delivery distance and fuel consumption because more choices are provided for route arrangement. To quantify the benefit, we define the benefit ratio between unshared and shared depots and prove that the maximum benefit ratios on route distance and fuel consumption are up to 2. Then the factors that affect the benefit ratio are analyzed by computational experiments. Computational results show that the benefit of shared depot resources depends on instance characteristics. Three characteristics are found to be significant to the relative performance, namely, depot–customer geographic distribution, maximum route distance, and number of depots.

1. Introduction

Governments and researchers have a growing awareness of the destruction caused by environmental pollution over the last decade. Carbon dioxide is the major contributor to the greenhouse effect. The International Energy Agency (2016) stated that the transportation industry is the second largest influence contributor of CO₂ emissions, accounting for 33% of the global CO₂ emissions in 2014, and almost 85% of the emissions from transportation are due to road transportation. Thus, fuel consumption and CO₂ emissions need to be reduced by optimizing vehicle routes in road transportation.

The environment-related vehicle routing problem (VRP) has received increasing interest (Xiao and Konak, 2015). This problem can be divided into three categories, namely, green VRP (GVRP), pollution-routing problem, and VRP in reverse logistics (Lin et al., 2014). The first category deals with the optimization of the energy consumption of transportation, including the analysis of fuel consumption models for road freight transportation (Demir et al., 2011; Demir et al., 2014a), and studies of VRP aimed at minimizing fuel consumption (Kuo, 2010; Xiao et al., 2012). The second category focuses on the reduction of pollution, especially carbon emissions (Bauer et al., 2010; Fagerholt et al., 2010). The third category concentrates on the distribution with reverse logistics because reuse and recycling are key processes in energy conservation and emission reduction (Kim et al., 2011; Ramos and Oliveira, 2011).

We focus on the multi-depot VRP with fuel consumption (MDVRPFC), which is related to the GVRP. In the classical multi-depot VRP (MDVRP), the route of each vehicle must end in the depot from which it starts. However, we consider a different variant of MDVRP, in which the vehicle is not required to return to the depot from which that it starts, and refer to this variant as an MDVRP under shared depot resources (MDVRPSDR). Our research motivation comes from E-commerce logistics in China. Recently, the E-commerce retail business in China is developing at a higher speed, with an annual growth of 29.4%, arriving at more than RMB

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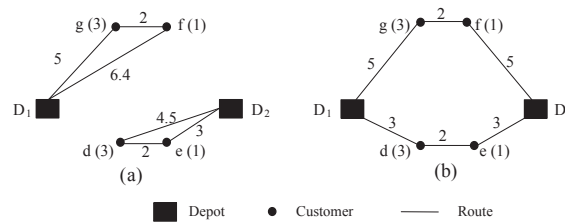


Fig. 1. Example for MDVRP under shared depot resources.

5 trillion in 2016. Multiple depots are established in large cities to serve consumers in time. For the traditional distribution centers, they only serve the customers within a fixed area. However, E-commerce logistics change the mode by the introduction of the Internet of things. Consequently, internet orders are flexibly assigned to different distribution centers, and the depot resources are shared among different distribution centers. The vehicles of a distribution center can share parking spaces of other distribution centers because drivers and vehicles can be managed by a global positioning system and an advanced information system. The drivers can go home through a convenient and fast subway system after they get off their works. Due to the pressures of increasing labor cost and fierce competition, shared depot resources are used to reduce the cost by some trucking enterprises. An example is shown in Fig. 1 and its notations are shown in Table 1. The optimal routes for the MDVRP are shown in Fig. 1(a), and those for the MDVRPSDR are shown in Fig. 1(b). In Fig. 1(a), the routes are D₁-g-f-D₁ and D₂-e-d-D₂, and their total length is 22.9. In Fig. 1(b) the routes are D₁-g-f-D₂ and D₂-e-d-D₁, and their total length is 20. The routes of the next round delivery are the same as those mentioned. The costs can be saved in MDVRPSDR because more choices for the route arrangement can be provided in MDVRPSDR than in MDVRP.

Recently, Demir et al. (2014c) indicated a logistics service provider could efficiently combine its resources to reduce its transport costs in a multi-depot network by planning the deliveries from multiple depots simultaneously. Markov et al. (2016) investigated the recyclable waste collection problem and translated it into the VRP with intermediate facilities. Except for the complex realistic features, vehicle tour could start from and end at different depots. Results showed that important savings were achieved in the state of practice. Li et al. (2016) studied the mathematical model and solution method of MDVRPSDR with the objective function of minimizing the travel distance. However, two unresolved problems remain, namely, (1) how to assess the maximum benefit under shared depot resources and (2) in which situation the benefit under shared depot resources is higher. No work has been done to study these two problems for MDVRPSDR.

Therefore, considering the two problems, we focus on the MDVRPSDR in the context of fuel consumption. The maximum potential benefit is evaluated by comparing the results under shared depot resources with those under unshared condition. The circumstance that is more beneficial for sharing depots is then analyzed.

The rest of this paper is structured as follows: Section 2 presents a literature review on the VRP with fuel consumption (VRPFC). Section 3 indicates the problem description. Section 4 provides the evaluation of the maximum benefit under shared depot resources. Section 5 provides the analysis of the factors that affect the benefit ratio between unshared and shared depots. Finally, Section 6 concludes the study and indicates the proposals for future research.

2. Literature review

In this study, we only investigate the VRPFC. For the broader review on green road freight transportation, the reader is referred to Lin et al. (2014) and Demir et al. (2014a). VRPFC is separated into two categories according to the number of depots, namely, single-depot VRPFC and MDVRPFC.

Considerable attention has been paid to the single-depot VRPFC. Xiao et al. (2012) developed a model for capacitated VRP, in which fuel consumption depends on load and distance. A simulated annealing (SA) algorithm with hybrid exchange rule was then designed to solve the problem. Yang et al. (2015) further examined VRP with time windows and multiple environmental vehicle types by using an HGA. The problem was formulated as a multi-objective model to minimize costs and environmental pollution and to maximize customer satisfaction. A sensitivity analysis was also conducted to assess the impact of different type vehicles on the environment. Kuo (2010) and Qian and Richard (2016) presented a time-dependent VRP, in which travel speeds varied with time. Their objective was to minimize fuel consumption. SA algorithm and column-generation-based tabu search algorithm were adopted, respectively, to identify the effect of speed on fuel consumption. Franceschetti et al. (2013) proposed a time-dependent pollution-

Table 1
Notations of the example.

Parameter	Notation
Depots	D ₁ and D ₂
Customers	d, e, f, and g
Arc length	The number on the arc
Demands of customer	The number in bracket
Capacity of vehicle	4

routing problem in which both vehicle speeds and emissions varied at different time periods. Their objective function considered traffic congestion. A novel departure time and speed optimization algorithm was then described to address the problem. Tajik et al. (2014) focused on determining optimal routes for pickup and delivery VRP with time window to minimize total costs, including transportation, time window violation penalty, CO₂ emission, and fuel consumption costs. They extended the previous problem, namely, fuel consumption and vehicle speed were defined varied with time under uncertainty. Soysal et al. (2015) addressed the time-dependent two-echelon capacitated VRP. The environmental impact was considered, and customers were served via intermediate depots in this problem. A real-life case study suggested that the two-echelon distribution system obtained a greener alternative, whereas the single-echelon distribution system obtained a less-cost solution. Zhang et al. (2015) considered a model setting in which both CO₂ emission and fuel consumption were incorporated into the objective function. An improved tabu search algorithm was then developed to solve numerical instances. Computational results showed that a positive correlation existed among distance, travel time, fuel consumption cost, and other parameters. Suzuki (2016) introduced six factors (speed, road gradient, congestion, driver, fleet size and mix, and payload) that impact fuel consumption according to the recent research efforts on VRPFC, and then developed a fuel consumption model by using only a subset of all these factors. Computational results showed that this model was practical.

Contrary to the single-depot situation, only a few papers on MDVRPFC exist. Chávez et al. (2016) proposed the MDVRP with backhauls by minimizing the costs of travel distance, travel time, and energy consumption, which was solved by ant colony algorithm. Ramos et al. (2014) constructed a multi-product and multi-depot VRP for recyclable waste collection systems in which two objectives of traveled distance and CO₂ emission were minimized. A real case study revealed that the program improved the economic and environmental benefits. However, research on the MDVRPFC under shared depot resources (MDVRPFCSDR) is lacking.

3. Problem description

We define the MDVRPFCSDR over a completely undirected graph $G = (V, E)$ with vertex set $V = \{v_1, \dots, v_N, v_{N+1}, \dots, v_{N+M}\}$ and arc set $E = \{(v_i, v_j) | v_i, v_j \in V, i \neq j\}$. The vertex is composed of two subsets V_c and V_d , where $V_c = \{v_1, \dots, v_N\}$ denotes N customers, and $V_d = \{v_{N+1}, \dots, v_{N+M}\}$ represents M depots. Each vertex $v_i \in V_c$ has a nonnegative demand q_i , whereas the demand of the depot vertex $D_i \in V_d$ is equal to zero. Two matrixes (distance matrix d_{ij} and speed matrix v_{ij}) are related to E . Finally, a fleet of vehicles with capacity Q exist at each depot $i \in V_d$. In the MDVRPFCSDR, the following constraints must be satisfied:

- (1) All vehicles are homogeneous.
- (2) Each vehicle must start from a depot and end at a depot after serving all customers, but the two depots are not necessarily the same.
- (3) The number of vehicles departing from each depot is not more than the number of available vehicles K_D .
- (4) The number of vehicles parking at a depot should not be more than the number of its parking spaces P_D .
- (5) The demand of each customer should be served exactly once by exactly one vehicle.
- (6) The total load of each vehicle does not exceed Q .
- (7) The travel distance of each route does not exceed T .

The objective of the problem is to determine the optimal routes by minimizing the total fuel consumption of all the vehicles. The objective function is formulated as follows,

$$\min H = \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_1 d_{ij} / v_{ij} + \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_2 d_{ij} (w + f_{kij}) x_{kij} + \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_3 v_{ij}^2 d_{ij} x_{kij} \quad (1)$$

where x_{kij} is 0–1 decision variable; if vehicle k travels directly from i to j , then $x_{kij} = 1$, otherwise, $x_{kij} = 0$. f_{kij} denotes load of vehicle k on arc (i, j) , w is curb weight of vehicle, and t_1 , t_2 , and t_3 are parameters, and K denotes vehicle set. Detailed mixed-integer linear programming formulation can be found in the Appendix A.

4. Evaluating the maximum benefit of shared depot resources

In this section, we aim to address the assessment of the maximum potential benefit under shared depot resources. The benefits of shared depot resources with the objective of minimizing the total travel distance are first assessed. The maximum fuel consumption savings under shared depot resources are then evaluated.

For multiple routes, benefits can be obtained by the benefits of one route multiplied by the number of routes. Therefore, we need to evaluate the maximum potential savings of one route. We discuss the situation of two depots, denoted by D_1 and D_2 , because each route can only connect two depots at most. Without loss of generality, the assumption is made that each vehicle starts from D_1 and ends at D_1 under unshared situation, and each vehicle departs from D_1 and ends at D_1 or D_2 under shared situation. Let us denote by l the last customer that is served under unshared depot condition, and by l' under shared depot condition. The minimum distance starting from depot D and visiting all customers in the set V_s , terminating at customer l , is expressed as $Z_l = Z(D, V_s, l)$, $V_s \subseteq V_c$. If the vehicle returns to the starting depot, then the minimum distance is $Z_{uns} = Z_{D_1} = Z(D_1, V_s, D_1) = \min_l [Z(D_1, V_s, l) + d_{lD_1}]$, $l \in V_s$; if the vehicle returns to the other depot, then the minimum distance is $Z_{ms} = Z_{D_2} = Z(D_1, V_s, D_2) = \min_{l'} [Z(D_1, V_s, l') + d_{l'D_2}]$, $l' \in V_s$. Therefore, the minimum travel distance under shared depot resources should be $Z_s = \min\{Z_{uns}, Z_{ms}\} = \begin{cases} Z_{uns} & \text{if } Z_{uns} < Z_{ms} \\ Z_{ms} & \text{otherwise} \end{cases}$. The

benefit ratio on route distance is denoted by $\frac{Z_{uns}}{Z_s}$.

Theorem 1. When the objective function is to minimize the traveling distance, $1 \leq \frac{Z_{uns}}{Z_s} \leq 2$.

Proof. We first consider that the last served customers are the same under unshared and shared depot situations, namely, $l = l'$. The optimal value with unshared depots is $Z_{uns} = Z_{l'} + d_{l'D_1}$, where $d_{l'D_1} \leq Z_{l'}$, and $d_{l'D_1}$ is equal to $Z_{l'}$ when the nodes visited by the route are in a straight line. When $Z_{ms} \leq Z_{uns}$ is satisfied, $0 \leq d_{l'D_2} < d_{l'D_1}$. Therefore, $\frac{Z_{uns}}{Z_s} = \frac{Z_{l'} + d_{l'D_1}}{Z_{l'} + d_{l'D_2}} \leq \frac{2Z_{l'}}{Z_{l'}} = 2$, and the bound is tight. When $d_{l'D_1} = Z_{l'}$ and $d_{l'D_2} = 0$ are true, namely, all customers in the route are in a straight line and the last visiting node is the depot D_2 , the ratio $\frac{Z_{uns}}{Z_s}$ is equal to 2.

We then consider the situation in which the last customer under unshared depot resources is different from the one under shared depot resources, namely, $l \neq l'$. In this situation, the objective function is to obtain the minimum value; thus, $Z(D_1, V_s, l) + d_{lD_1} \leq Z(D_1, V_s, l') + d_{l'D_1}$.

$$\frac{Z_{uns}}{Z_s} = \frac{Z(D_1, V_s, l) + d_{lD_1}}{Z(D_1, V_s, l') + d_{l'D_2}} \leq \frac{Z(D_1, V_s, l') + d_{l'D_1}}{Z(D_1, V_s, l') + d_{l'D_2}} \leq 2.$$

Thus, $\frac{Z_{uns}}{Z_s} \leq 2$ is necessarily true and $\frac{Z_{uns}}{Z_s} \geq 1$ always holds; $1 \leq \frac{Z_{uns}}{Z_s} \leq 2$ accordingly. $\square$

As in the preceding case, the minimum fuel consumption starting from depot d and visiting all customers in the set V_s , terminating at customer l , is expressed as $H_l = H(d, V_s, l), V_s \subseteq V_c$. If the vehicle returns to the starting depot, then the minimum fuel consumption is $H_{uns} = H_{D_1} = H(D_1, V_s, D_1) = \min_l [H(D_1, V_s, l) + F_{lD_1}], l \in V_s$; if the vehicle ends at the other depot, then the minimum fuel consumption is $H_{ms} = H_{D_2} = H(D_1, V_s, D_2) = \min_{l'} [H(D_1, V_s, l') + F_{l'D_2}], l' \in V_s$, where F_{ij} denotes vehicle fuel consumption traveling from customers i and j . Therefore, the minimum fuel consumption under shared depot resources should be $H_s = \min\{H_{uns}, H_{ms}\} = \begin{cases} H_{uns} & \text{if } H_{uns} < H_{ms} \\ H_{ms} & \text{otherwise} \end{cases}$.

The benefit ratio on fuel consumption is denoted by $\frac{H_{uns}}{H_s}$.

Theorem 2. Without considering the time-dependent speed, when the objective function is to minimize the fuel consumption, $1 \leq \frac{H_{uns}}{H_s} < 2$.

Proof. We initially consider that the last served customers are the same under unshared and shared depot situations, namely, $l = l'$. The minimum fuel consumption under shared depot resources is $H_{ms} = H_{l'} + F_{l'D_2}$, or the fuel consumption is $H_{uns} = H_{l'} + F_{l'D_1}$, where $F_{l'D_1} < H_{l'}$. When the nodes visited by the route are in a straight line, $d_{l'D_1}$ can be equal to $Z_{l'}$, but $F_{l'D_1}$ cannot be equal to $H_{l'}$ because the load of the vehicle is 0 when it travels back to the depot. When $H_{ms} \leq H_{uns}$ is true, $0 \leq F_{l'D_2} < F_{l'D_1}$. Therefore,

$$\frac{H_{uns}}{H_s} = \frac{H_{l'} + F_{l'D_1}}{H_{l'} + F_{l'D_2}} < \frac{2H_{l'}}{H_{l'}} = 2. \quad (2)$$

The situation in which the last customer under unshared depot resources is different from the one under shared depot resources, namely, $l \neq l'$, is then considered. In this situation, the objective function is to obtain the minimum value; thus, $H(D_1, V_s, l) + F_{lD_1} \leq H(D_1, V_s, l') + F_{l'D_1}$.

$$\frac{H_{uns}}{H_s} = \frac{H(D_1, V_s, l) + F_{lD_1}}{H(D_1, V_s, l') + F_{l'D_2}} \leq \frac{H(D_1, V_s, l') + F_{l'D_1}}{H(D_1, V_s, l') + F_{l'D_2}} < 2.$$

Therefore, $\frac{H_{uns}}{H_s} < 2$ is necessarily true without considering the time-dependent speed, and $\frac{H_{uns}}{H_s} \geq 1$ always holds. Consequently, $1 \leq \frac{H_{uns}}{H_s} < 2$. $\square$

The inequality (2) implies that the ratio of H_{uns} and H_s is closer to 2 when the value of $H_{l'}$ becomes smaller or the value of $F_{l'D_1}$ is closer to $H_{l'}$ or the vehicle load is decreased, or the vehicle travel distance is shortened.

5. Analysis on influence factors of the benefit ratio

In this section, we analyze how different factors affect the benefit under shared depot resources by computational experiments. The benchmark instances for MDVRP are modified to MDVRPFCSDR to assess different factor impacts on the benefit under shared depot resources. The hybrid genetic algorithm with adaptive local search (HGA_ALS) proposed by Li et al. (2016) is used to solve each instance under two scenarios, under both shared and unshared depots. The objective function in HGA_ALS is only replaced by formulation (1) to apply to MDVRPFCSDR. The effectiveness of HGA_ALS is assessed by testing on a set of standard MDVRP problem instances, in which the number of customers is no more than 100, shown in Section 5.1, while comparison of different factor impacts on benefit ratio is shown in Section 5.2.

5.1. Evaluating the performance of HGA_ALS on MDVRP instances

5.1.1. Experimental data

We evaluate the effectiveness of the HGA on 13 benchmark instances from Cordeau et al. (1995), in which the number of customers is no more than 100. In the instances, customers are generated by random uniform distribution in class A, by linear

Table 2
Characteristics of MDVRP instances.

Problem		MDVRP instances	N	M	T	Q
A	1	p01	50	4	∞	80
	2	p02	50	4	∞	160
	3	p03	75	5	∞	140
	4	p04	100	2	∞	100
	5	p05	100	2	∞	200
	6	P06	100	3	∞	100
	7	P07	100	4	∞	100
B	8	p12	80	2	∞	60
	9	p13	80	2	200	60
	10	p14	80	2	180	60
C	11	Pr01	48	4	500	200
	12	Pr02	96	4	480	195
	13	Pr07	72	6	500	200

distribution in class B, and by semi-clustered distribution in class C. The characteristics are shown in Table 2.

5.1.2. Comparison of results

The solutions obtained by the HGA and other state-of-the-art methodologies are shown in Table 3. Each instance is tested 10 times by the HGA. In Table 3, BKS denotes the best known solutions, AVG represents the average values of these 10 tests, and STD is the standard deviation. The last column reveals the gap between the best solution obtained by the HGA and the BKS, where gap % = (BEST–BKS)/BKS * 100.

As shown in Table 3, gaps% between the best solutions obtained by the HGA and BKS is 0.19% on average, with highest gap% at 1.02% and lowest gap% at 0.00%. The average value of 13 best results obtained by the HGA is better than that obtained by Pisinger and Ropk (2007) and Luo and Chen (2014). The maximum standard deviation is 2.84, and the average standard deviation is 0.71, which means that the HGA is relatively stable. The results show that the HGA is effective and competitive.

5.2. Comparison of different factor impacts on benefit ratio between MDVRPFC and MDVRPFCSDR

5.2.1 Experimental setup

Four factors, namely, number of depots, depot–customer geographic distribution, maximum route distance, and vehicle speed, are selected to assess the impacts on the benefit ratio $\lambda = H_{\text{uns}}/H_s$ to determine which situation would result in a high benefit ratio between unshared and shared depot scenarios. The values of parameters t_1 , t_2 , and t_3 in the fuel consumption objective function are the same as those reported by Demir et al. (2012), namely, $t_1 = 1.01763908$, $t_2 = 0.0000084$, $t_3 = 0.00014122$, and $w = 6350$. In addition, six instances with different customer sizes from 13 benchmark instances are used and redesigned for the study. They are denoted as I1, I2, I3, I4, I5, and I6. Each instance has two kinds of geographic distributions, namely, random uniform distribution (L_1) and linear distribution (L_2). Under the L_2 situation, customers are distributed near the connecting line between two depots. For the L_1 category, the original customer data are directly used; for the L_2 category, the coordinates of depots and partial customers are reset, and only the customer demands of benchmark instances are kept. The maximum route distance and vehicle speed are added to L_1 and L_2

Table 3
Comparison of results for MDVRP instances.

Problem	BKS	Pisinger and Ropk (2007)	Yu (2011)	Luo and Chen (2014)	HGA			Gap% to BKS
					BEST	AVG	STD	
1	576.86	576.87	576.86	576.87	576.87	576.87	0.00	0.00
2	473.53	473.53	473.53	473.53	473.53	473.53	0.00	0.00
3	640.65	641.19	641.18	641.19	641.19	641.66	1.09	0.08
4	999.21	1001.04	1001.49	1001.04	1002.23	1009.32	2.84	0.30
5	750.03	751.26	750.26	750.03	751.85	752.64	0.91	0.24
6	876.50	876.70	876.50	876.50	876.50	878.58	1.87	0.00
7	881.97	881.97	885.69	881.97	890.94	892.71	0.75	1.02
8	1318.95	1318.95	1318.95	1318.95	1318.96	1318.96	0.00	0.00
9	1318.95	1318.95	1318.95	1318.95	1318.96	1318.96	0.00	0.00
10	1360.12	1360.12	1365.68	1360.12	1360.12	1360.12	0.00	0.00
11	861.32	861.32	–	861.32	861.32	861.32	0.00	0.00
12	1296.25	1307.34	–	1307.34	1307.34	1308.87	1.76	0.86
13	1075.12	1089.56	–	1089.56	1075.12	1075.12	0.00	0.00
AVE	956.11	958.37	–	958.26	958.07	959.13	0.71	0.19

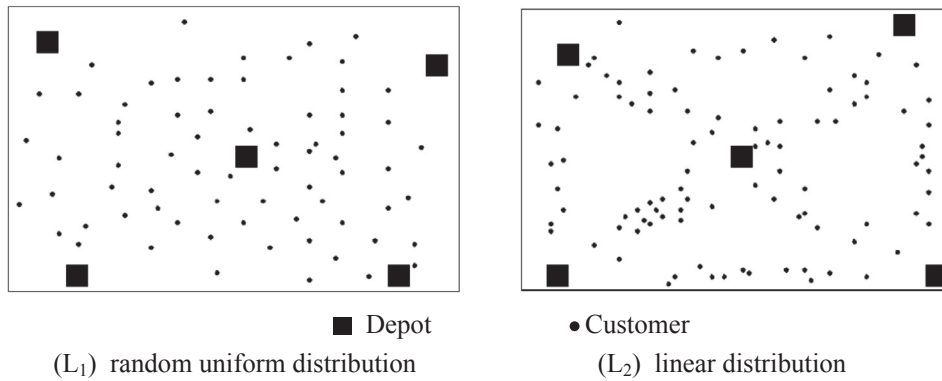


Fig. 2. Depot-customer geographic distribution.

categories. The numbers N of clients for the six instances are 50, 75, 100, 48, 96, and 72.

Depot-customer geographic distributions (L) and number of depots (M). The depot-customer geographic distribution can be divided into two categories, namely, L_1 and L_2 . These two distributions are used to estimate whether the depot-customer geographic distribution has an impact on the ratio λ . We set the number of depots as 4 and 5, denoted by K_1 and K_2 , respectively, to analyze whether the different numbers of depots affect the ratio λ . Examples of depot-customer geographic distributions are shown in Fig. 2, where five depots are included in the graph (when the number of depots is four, the middle depot in the graph is removed).

Vehicle speed (v). According to the fuel consumption model and parameter values, the optimum vehicle speed $v = 55$ km/h is obtained by taking the derivative of H with respect to speed v . For convenience of discussion, we assume that the speed of each arc in the graph is equal regardless of the time-dependent speed. Three different speeds, $v_1 = 20$ km/h, $v_2 = 30$ km/h, and $v_3 = 40$ km/h, are selected to test the instances and analyze the impact of speed on ratio λ .

Maximum route distance (T). In theorem 2, we determine that when F_{lD_2} is fixed, the value of H_l decreases, which means that the vehicle load decreases or the distance shortens (the distance must ensure that the depots can be shared), and the benefit ratio λ is closer to 2. Therefore, different maximum route distances are selected to test instances. The maximum distance between two depots is first defined and denoted by $T_{\max} = \max(d_{ij}), i, j \in V_d$ to quantify the maximum route distances. The depots can be shared only in the situation in which the route length is longer than the distance between depots. Thus, $T = a \times T_{\max}$ and $a \in \{1.2, 1.4, 1.6, 1.8\}$ are denoted, in which the values in a represents four-level maximum route distance, namely, $T_1 = 1.2T_{\max}$, $T_2 = 1.4T_{\max}$, $T_3 = 1.6T_{\max}$, and $T_4 = 1.8T_{\max}$.

The influence of four different factors on the benefit ratio λ is discussed under two scenarios of shared and unshared depots. The four parameters, including 11 levels, are shown in Table 4. Forty-eight ($2 \times 2 \times 3 \times 4$) combinations are examined for different levels of each parameter, and two states, where the depots are shared and unshared for each instance, are considered. Therefore, each instance needs to be tested for 96 (48×2) times. In this study, six instances are considered; thus, 576 (96×6) tests are executed. The value of λ is used to estimate the savings of shared depots compared with unshared depots. The larger the value of λ is, the more the savings of sharing depots are. All detailed test results can be found in the Appendix B.

5.2.2. Computational results

Table 5 reports the relative performance results of fuel consumption under shared and unshared depots. This table can be divided into four groups with two to four different levels according to influence factors. Each row provides the relative performance λ by fixing each parameter to a specific level and varying all other parameters. For instance, 1.05 in the first cell of Table 5 represents the average value of 24 ($2 \times 3 \times 4$) combinations in instance I1 with L_1 level. The influence of each factor on the benefit ratio λ is analyzed in detail in the following.

Depot-customer geographic distribution (L). The results of six instances are depicted in Fig. 3, which graphically describes the results obtained in the first group in Table 5. The average values in Table 5 imply that λ is increased by 6.8% (from 1.03 to 1.10) when the depot-customer distribution level moves from L_1 to L_2 , which is the most significant growth among all parameters. Therefore, the depot-customer geographic distribution affects the fuel consumption significantly. The overall average value of the benefit ratio λ reaches the maximum when customers are distributed near the straight lines between depots.

Table 4
Values of 11 levels of four parameters.

Level/Parameter	Depot-customer distribution(L)	Number of depots(M)	Vehicle speed(v)	Maximum route distance(T)
1	Random uniform distribution	4	20 km/h	$1.2 \times T_{\max}$
2	Linear distribution	5	30 km/h	$1.4 \times T_{\max}$
3	—	—	40 km/h	$1.6 \times T_{\max}$
4	—	—	—	$1.8 \times T_{\max}$

Table 5
Average λ values.

Level/Instance	I1	I2	I3	I4	I5	I6	Average
L_1	1.05	1.02	1.02	1.03	1.03	1.04	1.03
L_2	1.10	1.11	1.11	1.11	1.10	1.09	1.10
M_1	1.05	1.05	1.06	1.06	1.06	1.05	1.06
M_2	1.11	1.08	1.08	1.09	1.07	1.08	1.09
v_1	1.08	1.06	1.07	1.07	1.07	1.07	1.07
v_2	1.08	1.07	1.07	1.07	1.07	1.07	1.07
v_3	1.08	1.07	1.07	1.07	1.07	1.07	1.07
T_1	1.10	1.08	1.11	1.09	1.10	1.10	1.10
T_2	1.08	1.07	1.07	1.07	1.08	1.05	1.07
T_3	1.08	1.05	1.05	1.07	1.05	1.06	1.06
T_4	1.06	1.05	1.04	1.06	1.05	1.06	1.05
Average	1.08	1.06	1.07	1.07	1.07	1.07	1.07

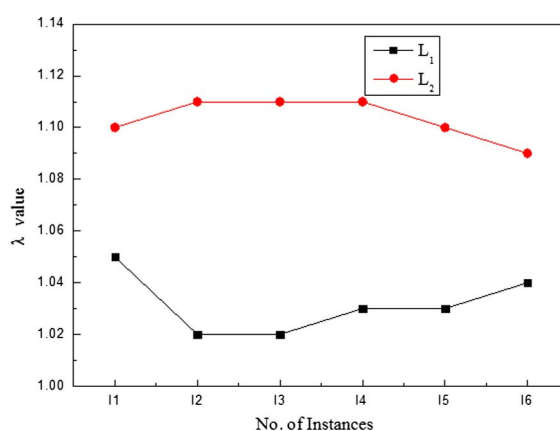


Fig. 3. Impact of depot–customer distribution on λ value.

A new instance I7 is designed to further reflect the effect of depot–customer distribution on the ratio λ . The depot–customer geographic distribution in instance I7 is depicted in Fig. 4. In Table 6, the results in columns 3 and 4 are the average values of three vehicle speed levels. The fuel consumption under shared depots is obviously saved compared with that under an unshared depot. The average λ value is 1.32 when customers are located on the straight line between depots, as shown in Table 6. Thus, if customers are in the connection lines between depots, then sharing depots becomes favorable.

Maximum route distance (T). The proof in theorem 2 shows that the shorter route distance leads to the greater λ ratio, under the premise that sharing depots is allowed. The relative performance of six instances is depicted in Fig. 5. Moving from T_4 to T_1 in the Table 5, the route distance reduction yields a 4.76% average increase on λ value (from 1.05 to 1.10), which presents a gradual upward trend. This trend is shown based on a statistical regularity. The test results for each case fluctuate because they are affected by vehicle speed and customer distribution. However, such a conclusion turns out to be true on one condition, i.e., the route distance must be set within a reasonable range. A short route that is even shorter than the distance between two depots will limit the

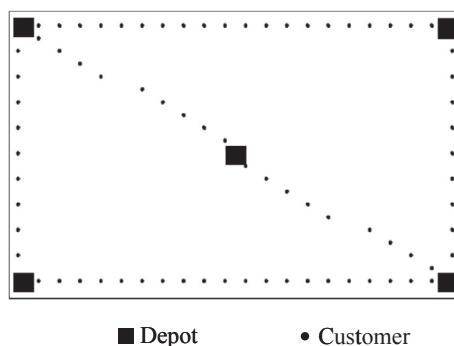


Fig. 4. Depot–customer distribution of instance I7.

Table 6
Fuel consumption results of I7 with L_2 .

Number of depots	Level	H_{uns}	H_s	λ
4 depots	T_1	228.07	167.91	1.36
	T_2	219.92	164.96	1.33
	T_3	211.65	164.96	1.28
	T_4	201.51	162.82	1.24
5 depots	T_1	226.03	164.01	1.38
	T_2	217.34	164.02	1.33
	T_3	215.39	161.66	1.33
	T_4	207.46	161.66	1.28
Average		215.92	164.00	1.32

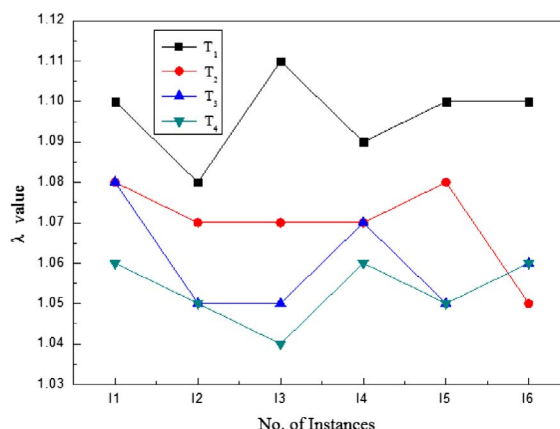


Fig. 5. Impact of maximum route distance on λ value.

possibility of sharing, consequently causing a decrease in λ value. For validation, $T_{\min} = \min(d_{ij}), i, j \in V_d$ is defined to represent the minimum distance between two depots. Instance I1 is then further tested by adding the constraint that the distance should be less than $T_{\max}$ but greater than $T_{\min}$. Thus, four more distance levels are used, namely, $T_a = T_{\min}$, $T_b = 1.2 \times T_{\min}$, $T_c = 1.4 \times T_{\min}$, and $T_d = T_{\max}$. As shown in Table 7, the test results are the values at speed level v_3 .

The test results show that when the maximum route distance is equal to T_a , the saving of sharing depots cannot be achieved. In this situation, the fuel consumptions of shared and unshared depots are the same, thereby resulting in the value of λ equal to 1. When the maximum route distance is equal to T_b and T_c , the advantage of shared depots on fuel consumption is gradually revealed because a few depots can share with other depots. When the maximum route distance changes from T_c to T_d , the ratio λ of two types of depot–customer geographic distribution increases obviously (from 1.01 to 1.09 and from 1.02 to 1.11). If the route distance continues to increase, then the value of λ decreases after reaching its maximum. Therefore, when the route distance moves from T_a level to T_4 level, the benefit ratio λ first increases from 1 to the maximum and then decreases, as shown in Fig. 6.

Another observation from Table 6 is that the best route distance will make the advantages of shared depots more obvious under customer linear distribution. In instance I7, when the nodes are linearly distributed and the maximum route distance is at T_1 level, the

Table 7
Fuel consumption results of instance I1.

Number of depots	Level	I1- L_1			I1- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
5 depots	T_a	211.16	211.16	1.00	203.27	203.27	1.00
	T_b	171.64	170.87	1.01	175.99	174.87	1.01
	T_c	139.59	138.34	1.01	148.16	146.03	1.02
	T_d	95.88	88.06	1.09	96.20	86.48	1.11
	T_1	84.20	75.20	1.12	80.30	69.20	1.16
	T_2	76.60	70.80	1.08	73.20	64.40	1.14
	T_3	73.20	69.00	1.06	70.00	62.20	1.13
	T_4	70.90	67.50	1.05	66.30	60.10	1.10
Average		115.40	111.37	1.05	114.18	108.32	1.08

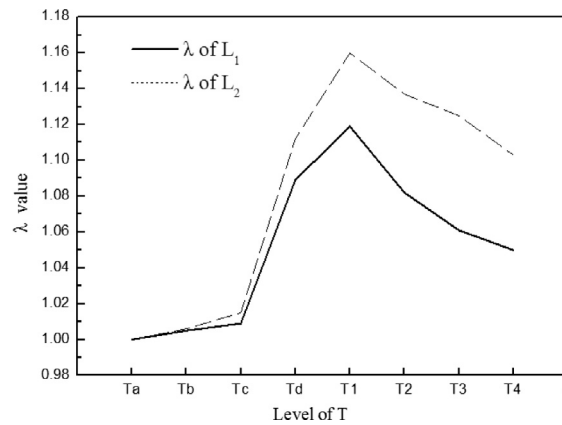


Fig. 6. λ values of instance I1 with different maximum route distances.

ratio λ reaches 1.36 and 1.38 under situations that the numbers of depots are 4 and 5, respectively. The fuel consumptions under unshared depot are accordingly more than 36% and 38% of those under shared depots.

Number of depots (M). The relative performance of six instances is depicted in Fig. 7. When the number of depots moves from M_1 to M_2 level, the λ values increase. When the quantity of depots increases, the opportunity for the shared depot is increased because of the increased number of connection lines between depots. Therefore, when a larger number of depots are available, the benefit ratio λ increases.

Vehicle speed (v). Although the vehicle speed has an impact on fuel consumption, the change in fuel consumption under shared and unshared depots is the same at different speed levels, as shown in Table 5. Therefore, the average λ values remain stable at different vehicle speeds.

The MDVRPFC under shared depot is superior to the condition with unshared depots. All λ in instances implies that the minimum benefit ratio is 1.00, and the maximum value is up to 1.38. The preceding analysis reveals that the benefits of sharing depots are obvious in the following three situations: (1) customer distribution near the connecting lines between the depots, (2) moderate maximum route distance, and (3) larger number of depots. The computational results showed that the depot–customer distribution has the most important impact on the value of λ among the performances of all factors. The second important factor is the maximum route distance.

6. Conclusions

In this study, the maximum potential benefit under shared depot resources was evaluated through comparison with the unshared condition. It was proved that the maximum benefit ratios on route distance and fuel consumption were 2, but the benefit ratio for the former can be equal to 2, and strictly less than 2 for the latter. Four factors that affect the benefit ratio were then analyzed by computational experiments. Computational results showed three of them, namely, depot–customer geographic distribution, maximum route distance, and number of depots, were closely related to the value of benefit ratio, in which the depot–customer geographic distribution affected the benefit ratio most obviously. By contrast, vehicle speed was not significant with the benefit ratio. Based on analyses of three factors which closely affect the benefit ratio, we give the following recommendations to logistics

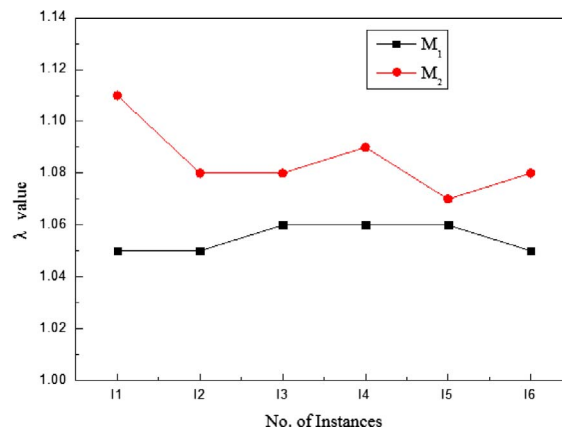


Fig. 7. Impact of depot number on λ value.

managers:

- (1) The shared depot resources can obtain more saving if the customers distribute near the connect line between warehouses, for example, customers are located along the streets between warehouses.
- (2) For enterprises with online-offline platform, they have many online customers and offline customers. Their retail stores can have the functions of store and warehouse, thus the online customers can be served by retail stores. In this situation, these enterprises will have more distribution depots, and shared depot is more beneficial for them.
- (3) The shared depot is also more beneficial to perishable and fresh food industry, because they give short time to transport, which means the maximum route distance is not longer.

For future research, one interesting direction is to extend the sharing resources, such as manpower resources.

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Appendix A

A.1. Notations

Sets	
V_d	depot set
V_c	customer set
V	vertex set
E	arc set
K	vehicle set
Parameters	
q_i	demand of customer i
d_{ij}	distance between customers i and j
v_{ij}	vehicle speed between i and j
w	curb weight of vehicle, i.e., weight of an empty vehicle
Q	maximum capacity of a vehicle
T	maximum route distance
K_D	number of vehicles in depot D
P_D	number of parking spaces in depot D
Variables	
f_{kij}	load of vehicle k on arc (i,j)
x_{kij}	0–1 decision variable; if vehicle k travels directly from i to j , then $x_{kij} = 1$; otherwise, $x_{kij} = 0$

A.2. Mixed-integer linear programming formulation

We use the fuel consumption model described by Demir et al.(2012, 2014b), which is an extension of the model by Barth et al. (2005) and Barth and Boriboonsomsin (2008), to calculate fuel consumption, considering the influences of vehicle load, travel distance, and travel speed on fuel consumption. The fuel consumption model is constructed as

$$F(d,v,f) = FR \times (d/v), \quad (A1)$$

where F denotes fuel consumption, d is distance (meter), V represents vehicle speed (meter/s), and d/v represents time. FR is the fuel rate that can be specifically expressed as

$$FR = \xi(\mu B \delta + (P_{tract}/\eta_f + P_{acc})/\eta)/\kappa. \quad (A2)$$

Formula (A2) is designed to calculate the fuel consumption generated in unit time, where $\mu, B, \delta, \eta_f, P_{acc}$ are the parameters related to the vehicle equipment, namely, engine friction factor, engine speed, engine displacement, vehicle drive train efficiency, and engine power demand, respectively. P_{acc} is usually assumed to be zero (Demir et al., 2012). P_{tract} represents the total traction power (kilowatt) required for wheels. This parameter is related to the vehicle state during transit operation and the road condition, which may be expressed as

$$P_{tract} = (M\tau + Mgsin\theta + 0.5C_d\rho Av^2 + MgC_r cos\theta)v/1000, \quad (A3)$$

where τ, C_d, C_r are the parameters related to the motoring condition, which are the acceleration, coefficient of aerodynamic drag, and rolling resistance, respectively. θ is related to the geographic feature of the distribution area, which indicates the road gradient. A is a vehicle-related parameter. $\eta, \kappa, \rho, g, \xi$ are constants that signify the efficiency parameter for diesel engines, heating value of a typical diesel fuel, air density, gravitational constant, and fuel-to-air mass ratio, respectively. M , which denotes the entire weight of the vehicle (kilogram), can be calculated as $M = w + f$, where w is the curb weight of the vehicle, and f is the loading weight of the vehicle; all these are factors of fuel consumption. Therefore, the fuel consumption for vehicle can be expressed as

$$F(d, v, f) = \frac{\xi}{\kappa\psi} \mu BV (d/v) + \frac{\xi}{\kappa\psi} \times \frac{1}{1000\eta_{ff}\eta} (\tau + g \sin \theta + g C_r \cos \theta) \times d(w + f) + \frac{\xi}{\kappa\psi} \times \frac{1}{1000\eta_{ff}\eta} \times 0.5 C_d \rho A \times dv^2. \quad (A4)$$

Then, the constant parameter of Formula (A4) is merged. When a vehicle travels on arc (i, j) at speed v with load f , the fuel consumption (liter) model can be simply denoted as

$$F(d, v, f) = t_1 d_{ij}/v_{ij} + t_2 d_{ij}(w + f_{ij}) + t_3 d_{ij} v_{ij}^2, \quad (A5)$$

where $t_1 = \frac{\xi}{\kappa\psi} \mu BV$, $t_2 = \frac{\xi}{\kappa\psi} \times \frac{1}{1000\eta_{ff}\eta} (\tau + g \sin \theta + g C_r \cos \theta)$, and $t_3 = \frac{\xi}{\kappa\psi} \times \frac{1}{1000\eta_{ff}\eta} \times 0.5 C_d \rho A$; all of these are constants.

According to the preceding discussion, the model for MDVRPFCSDR should be formulated as follows:

$$\min H = \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_1 d_{ij}/v_{ij} + \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_2 d_{ij}(w + f_{kij})x_{kij} + \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} t_3 v_{ij}^2 d_{ij}x_{kij} \quad (A6)$$

S.t.

$$\sum_{k \in K} \sum_{j \in V_c} x_{kDj} \leq K_D, \forall D \in V_d \quad (A7)$$

$$\sum_{k \in K} \sum_{i \in V_c} (x_{kiD} - x_{kDi}) \leq P_D - K_D, \forall D \in V_d \quad (A8)$$

$$\sum_{k \in K} \sum_{j \in V} x_{kij} = \sum_{k \in K} \sum_{j \in V} x_{kji} = 1, \forall i \in V_c \quad (A9)$$

$$\sum_{D \in V_d} \sum_{j \in V_c} x_{kDj} = \sum_{D \in V_d} \sum_{i \in V_c} x_{kiD} \leq 1, \forall k \in K \quad (A10)$$

$$\sum_{i \in C} \sum_{j \in C} x_{kij} \leq |C| - 1, \forall C \subseteq V_c, k \in K \quad (A11)$$

$$\sum_{i \in V_d} \sum_{j \in V_d} x_{kij} = 0, \forall k \in K \quad (A12)$$

$$\sum_{k \in K} \sum_{j \in V} f_{kji} - \sum_{k \in K} \sum_{j \in V} f_{kij} = q_i, \forall i \in V_c \quad (A13)$$

$$\sum_{i \in V} \sum_{j \in V_c} q_j x_{kij} \leq Q, \forall k \in K \quad (A14)$$

$$\sum_{i \in V} \sum_{j \in V} d_{ij} x_{kij} \leq T, \forall k \in K \quad (A15)$$

$$f_{kij} \geq 0, \forall i, j \in V, k \in K \quad (A16)$$

$$x_{kij} \in \{0, 1\}, \forall (i, j) \in E, k \in K \quad (A17)$$

The objective function (A6) minimizes the fuel consumption, in which the first item and third item denote the fuel consumption caused by the vehicle speed and the second item denote the fuel consumption caused by the empty vehicle and the loading weight of the vehicle. Constraint (A7) indicates that the number of vehicles used in each depot is not more than the number of available vehicles. Constraint (A8) represents that the number of vehicles parked in each depot cannot exceed the number of available parking places. Constraint (A9) requires that each customer be serviced once by only one vehicle. Constraint (A10) means that each vehicle must start from a depot and end at a depot. Constraint (A11) presents the subtour elimination. Constraint (A12) ensures the vehicle cannot travel directly from depot to depot. Constraint (A13) indicates the constraint about flow conservation; every time a customer is visited, its demand quantity remains equal to the difference between the load before serving the customer and the load after serving the customer. Constraint (A14) represents the vehicle capacity constraints. Constraint (A15) denotes the maximum route distance constraints. Constraint (A16) indicates that the loading weight of the vehicle has a nonnegative attribute. Constraint (A17) specifies the range of the decision variable.

Appendix B

See Tables B1–B7.

Table B1

The fuel consumption results of instance I1.

Number of depots	Level	I1- L_1			I1- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	77.20	75.90	1.02	75.10	68.80	1.09
	T_1, v_2	88.10	86.00	1.02	85.60	78.50	1.09
	T_1, v_1	114.33	112.35	1.02	111.15	101.85	1.09
	T_2, v_3	71.70	70.30	1.02	68.60	64.50	1.06
	T_2, v_2	81.80	80.20	1.02	78.30	73.60	1.06
	T_2, v_1	106.18	104.12	1.02	101.56	95.50	1.06
	T_3, v_3	71.20	69.40	1.03	67.60	62.00	1.09
	T_3, v_2	81.20	79.20	1.03	77.10	70.70	1.09
	T_3, v_1	105.43	103.25	1.02	100.00	91.67	1.09
	T_4, v_3	68.40	66.80	1.02	64.10	61.00	1.05
	T_4, v_2	78.00	76.40	1.02	73.10	69.60	1.05
	T_4, v_1	101.21	98.90	1.02	94.80	90.20	1.05
5 depots	T_1, v_3	84.20	75.20	1.12	80.30	69.20	1.16
	T_1, v_2	96.10	85.80	1.12	91.60	78.90	1.16
	T_1, v_1	124.73	111.43	1.12	118.94	102.50	1.16
	T_2, v_3	76.60	70.80	1.08	73.20	64.40	1.14
	T_2, v_2	88.71	80.80	1.10	83.50	73.40	1.14
	T_2, v_1	113.44	104.85	1.08	108.38	95.35	1.14
	T_3, v_3	73.20	69.00	1.06	70.00	62.20	1.13
	T_3, v_2	83.50	78.50	1.06	79.80	70.90	1.13
	T_3, v_1	108.46	101.48	1.07	103.60	92.05	1.13
	T_4, v_3	70.90	67.50	1.05	66.30	60.10	1.10
	T_4, v_2	80.90	77.06	1.05	75.60	68.50	1.10
	T_4, v_1	104.97	100.00	1.05	98.10	89.00	1.10

Table B2

The fuel consumption results of instance I2.

Number of depots	Level	I2- L_1			I2- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	104.72	102.13	1.03	98.95	87.90	1.13
	T_1, v_2	119.44	116.49	1.03	112.86	99.41	1.14
	T_1, v_1	155.02	151.22	1.03	146.48	129.00	1.14
	T_2, v_3	96.78	96.78	1.00	93.40	84.42	1.11
	T_2, v_2	110.35	110.35	1.00	106.51	96.30	1.11
	T_2, v_1	143.09	143.09	1.00	138.13	124.92	1.11
	T_3, v_3	94.40	92.70	1.02	84.81	81.20	1.04
	T_3, v_2	107.86	106.19	1.02	96.70	92.60	1.04
	T_3, v_1	139.58	138.09	1.01	125.42	120.07	1.04
	T_4, v_3	90.40	90.40	1.00	84.46	80.40	1.05
	T_4, v_2	102.59	102.46	1.00	96.40	91.80	1.05
	T_4, v_1	133.22	133.22	1.00	124.81	119.19	1.05
5 depots	T_1, v_3	101.75	100.82	1.01	100.34	87.30	1.15
	T_1, v_2	116.07	115.19	1.01	114.45	99.61	1.15
	T_1, v_1	150.69	148.86	1.01	148.57	129.30	1.15
	T_2, v_3	96.30	93.10	1.03	95.70	83.00	1.15
	T_2, v_2	109.86	106.07	1.04	109.20	94.60	1.15
	T_2, v_1	142.55	137.72	1.04	141.63	122.81	1.15
	T_3, v_3	93.50	90.30	1.04	90.60	79.90	1.13
	T_3, v_2	106.58	103.02	1.03	103.30	91.90	1.12
	T_3, v_1	138.28	133.86	1.03	134.05	118.31	1.13
	T_4, v_3	90.20	88.00	1.03	86.86	77.40	1.12
	T_4, v_2	103.07	100.41	1.03	99.10	88.30	1.12
	T_4, v_1	134.96	130.86	1.03	128.52	115.72	1.11

Table B3

The fuel consumption results of instance I3.

Number of depots	Level	I3- L_1			I3- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	123.15	117.67	1.05	113.95	99.35	1.15
	T_1, v_2	139.23	134.86	1.03	129.99	113.21	1.15
	T_1, v_1	182.21	176.38	1.03	167.99	146.91	1.14
	T_2, v_3	114.48	111.71	1.02	105.20	93.90	1.12
	T_2, v_2	130.06	128.05	1.02	121.91	107.04	1.14
	T_2, v_1	169.92	166.00	1.02	154.98	138.89	1.12
	T_3, v_3	108.95	108.10	1.01	95.40	89.50	1.07
	T_3, v_2	124.32	123.06	1.01	107.77	102.99	1.05
	T_3, v_1	161.20	160.69	1.00	139.65	132.26	1.06
	T_4, v_3	107.31	107.31	1.00	93.20	89.20	1.04
	T_4, v_2	122.35	122.35	1.00	107.48	102.82	1.05
	T_4, v_1	158.73	158.70	1.00	137.81	131.87	1.05
5 depots	T_1, v_3	122.93	115.80	1.06	112.81	94.17	1.20
	T_1, v_2	140.55	133.05	1.06	128.68	107.46	1.20
	T_1, v_1	181.68	170.92	1.06	167.03	139.41	1.20
	T_2, v_3	111.64	111.13	1.00	103.29	92.40	1.12
	T_2, v_2	127.61	126.88	1.01	117.80	105.29	1.12
	T_2, v_1	165.36	164.77	1.00	155.90	136.42	1.14
	T_3, v_3	109.05	107.10	1.02	100.05	89.60	1.12
	T_3, v_2	124.37	121.98	1.02	114.10	102.08	1.12
	T_3, v_1	161.41	158.01	1.02	148.05	132.47	1.12
	T_4, v_3	107.23	106.09	1.01	96.30	87.20	1.10
	T_4, v_2	122.67	121.06	1.01	108.32	100.20	1.08
	T_4, v_1	159.03	156.99	1.01	140.66	130.00	1.08

Table B4

The fuel consumption results of instance I4.

Number of depots	Level	I4- L_1			I4- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	161.97	151.69	1.07	138.96	126.01	1.10
	T_1, v_2	184.91	173.06	1.07	158.52	143.75	1.10
	T_1, v_1	242.31	224.7	1.08	205.81	186.63	1.10
	T_2, v_3	145.34	144.54	1.01	129.38	118.75	1.09
	T_2, v_2	165.8	164.89	1.01	147.58	135.46	1.09
	T_2, v_1	215.26	214.07	1.01	191.58	175.84	1.09
	T_3, v_3	139.32	138.14	1.01	125.75	115.42	1.09
	T_3, v_2	158.93	157.57	1.01	143.43	132.98	1.08
	T_3, v_1	206.32	204.56	1.01	186.18	170.98	1.09
	T_4, v_3	129.97	129.86	1.00	120.96	112.96	1.07
	T_4, v_2	148.24	148.12	1.00	137.97	128.83	1.07
	T_4, v_1	192.38	192.24	1.00	179.08	167.78	1.07
5 depots	T_1, v_3	154.29	148.59	1.04	141.62	124.69	1.14
	T_1, v_2	176.05	169.53	1.04	161.57	142.26	1.14
	T_1, v_1	228.62	220.15	1.04	209.82	184.71	1.14
	T_2, v_3	142.86	139.69	1.02	135.46	117.61	1.15
	T_2, v_2	162.99	159.37	1.02	154.55	134.17	1.15
	T_2, v_1	211.63	206.91	1.02	200.68	174.19	1.15
	T_3, v_3	139.87	133.34	1.05	128.37	111.45	1.15
	T_3, v_2	159.57	152.07	1.05	146.44	127.13	1.15
	T_3, v_1	207.19	197.42	1.05	190.12	165.03	1.15
	T_4, v_3	133.87	127.78	1.05	124.48	109.98	1.13
	T_4, v_2	152.73	145.77	1.05	141.99	125.46	1.13
	T_4, v_1	198.28	189.24	1.05	184.32	162.85	1.13

Table B5
The fuel consumption results of instance I5.

Number of depots	Level	I5- L_1			I5- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	229.72	225.44	1.02	173.07	148.82	1.16
	T_1, v_2	262.05	257.02	1.02	197.39	169.81	1.16
	T_1, v_1	340.17	334.55	1.02	256.20	220.26	1.16
	T_2, v_3	214.11	208.93	1.02	161.81	146.32	1.11
	T_2, v_2	244.17	238.43	1.02	183.68	166.91	1.10
	T_2, v_1	316.89	310.86	1.02	239.14	216.47	1.10
	T_3, v_3	208.56	203.60	1.02	153.88	144.04	1.07
	T_3, v_2	237.76	232.23	1.02	175.47	164.23	1.07
	T_3, v_1	308.58	301.30	1.02	227.64	213.09	1.07
	T_4, v_3	203.78	198.07	1.03	151.36	142.59	1.06
	T_4, v_2	229.61	225.89	1.02	172.47	162.35	1.06
	T_4, v_1	297.17	293.17	1.01	223.69	208.70	1.07
5 depots	T_1, v_3	233.25	220.12	1.06	170.60	149.12	1.14
	T_1, v_2	266.16	251.13	1.06	194.55	170.13	1.14
	T_1, v_1	345.62	326.06	1.06	252.60	220.89	1.14
	T_2, v_3	218.02	206.57	1.06	164.05	145.82	1.13
	T_2, v_2	248.52	235.69	1.05	187.00	165.96	1.13
	T_2, v_1	322.64	305.98	1.05	242.74	215.37	1.13
	T_3, v_3	202.99	199.37	1.02	153.88	143.99	1.07
	T_3, v_2	231.59	225.64	1.03	175.54	163.99	1.07
	T_3, v_1	300.56	292.96	1.03	227.81	212.78	1.07
	T_4, v_3	197.02	195.87	1.01	152.49	140.03	1.09
	T_4, v_2	224.73	223.32	1.01	173.93	159.31	1.09
	T_4, v_1	291.74	289.89	1.01	225.74	204.23	1.11

Table B6
The fuel consumption results of instance I6.

Number of depots	Level	I6- L_1			I6- L_2		
		H_{uns}	H_s	λ	H_{uns}	H_s	λ
4 depots	T_1, v_3	195.95	186.34	1.05	160.43	144.44	1.11
	T_1, v_2	223.54	212.55	1.05	183	164.7	1.11
	T_1, v_1	290.23	275.92	1.05	237.57	213.75	1.11
	T_2, v_3	185.3	180.04	1.03	147.85	140.41	1.05
	T_2, v_2	211.36	202.71	1.04	168.62	160.65	1.05
	T_2, v_1	274.36	262.86	1.04	218.83	207.76	1.05
	T_3, v_3	179.12	174.41	1.03	145.83	138.59	1.05
	T_3, v_2	203.75	198.39	1.03	166.59	155.35	1.07
	T_3, v_1	265.18	257.6	1.03	216.18	203.83	1.06
	T_4, v_3	170.81	170.06	1.00	142.97	133.46	1.07
	T_4, v_2	194.82	193.96	1.00	162.14	153.76	1.05
	T_4, v_1	252.85	248.86	1.02	210.36	196.56	1.07
5 depots	T_1, v_3	206.56	192.44	1.07	169.36	146.37	1.16
	T_1, v_2	235.7	219.51	1.07	193.15	166.98	1.16
	T_1, v_1	306.02	285.14	1.07	250.77	216.8	1.16
	T_2, v_3	187.57	180.19	1.04	152.97	140.03	1.09
	T_2, v_2	213.94	205.62	1.04	174.43	159.75	1.09
	T_2, v_1	277.8	268.43	1.03	226.43	207.4	1.09
	T_3, v_3	184.68	175.03	1.06	147.26	134.02	1.10
	T_3, v_2	210.68	199.59	1.06	167.97	152.87	1.10
	T_3, v_1	273.52	259.13	1.06	218.02	198.42	1.10
	T_4, v_3	179.24	169.32	1.06	145.16	131.72	1.10
	T_4, v_2	204.49	194.77	1.05	165.58	150.33	1.10
	T_4, v_1	265.51	250.84	1.06	214.92	194.92	1.10

Table B7

The fuel consumption results of instance I7.

Number of depots	Level	I7-L ₂		
		H_{uns}	H_s	λ
4 depots	T_1, v_3	188.89	139.06	1.36
	T_1, v_2	215.50	158.66	1.36
	T_1, v_1	279.82	206.01	1.36
	T_2, v_3	182.15	136.65	1.33
	T_2, v_2	207.81	155.88	1.33
	T_2, v_1	269.81	202.35	1.33
	T_3, v_3	175.30	136.65	1.28
	T_3, v_2	199.99	155.88	1.28
	T_3, v_1	259.65	202.35	1.28
	T_4, v_3	166.92	134.91	1.24
	T_4, v_2	190.41	153.87	1.24
	T_4, v_1	247.19	199.67	1.24
5 depots	T_1, v_3	187.18	135.84	1.38
	T_1, v_2	213.56	154.98	1.38
	T_1, v_1	277.34	201.22	1.38
	T_2, v_3	180.00	135.84	1.33
	T_2, v_2	205.37	154.98	1.33
	T_2, v_1	266.66	201.24	1.33
	T_3, v_3	178.39	133.90	1.33
	T_3, v_2	203.52	152.76	1.33
	T_3, v_1	264.25	198.31	1.33
	T_4, v_3	171.84	133.91	1.28
	T_4, v_2	196.11	152.76	1.28
	T_4, v_1	254.42	198.31	1.28

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